PHYS 570: Quantum Biology

Lecture 01a

Crash intro to quantum mechanics

De Broglie wavelength
Time-dependent Schrödinger equation (TDSE)
Time-independent Schrödinger equation (TISE)
Free particle, particle in a box, particle in a harmonic potential
Operators, measurements, expectation values
Operators, eigenvalues, and eigenfunctions
Matrix representation of operators and wavefunctions

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Matter: wave-particle duality

Louis de Broglie
1910: Sorbonne, his first degree in history
1913: degree in science (influenced by his brother)
1914: military service (World War I)
1919: continued scientific studies

Louis Victor Pierre Raymond, 7th Duc de Broglie 1892-1987

Idea:
Photon energy is: $E = h\nu = hc/\lambda$
if it is a particle, according to Einstein: $E = mc^2$

$$\Rightarrow \lambda = h/mc = h/p$$

→ Extend this idea to other objects, like electrons, with momentum *p=mv*

de Broglie wavelength

$$\lambda = \frac{h}{p} = \frac{h}{mv}$$

Frequency

$$\nu = \frac{E}{h}$$

Note: a more elaborate derivation can be done starting from the group velocity of a wave packet, $v = d\omega/dk$, $h\nu = \text{sqrt}(p^2c^2+m_0^2c^2)$
See https://physics.stackexchange.com/questions/223720/doubt-regarding-de-broglie-relation-from-group-velocity

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De Broglie wavelength

de Broglie wavelength

$$\lambda = \frac{h}{p} = \frac{h}{mv}$$

Frequency

$$\nu = \frac{E}{h}$$

Particle		**Wave**
Energy	⟷	Frequency
Momentum	⟷	Wavelength

Explained Bohr's model of the atom (standing waves)

2024, Thesis « Recherches sur la Théorie des Quanta » (Researches on the quantum theory)
(defended at the Faculty of Sciences at Paris University)

(Nobel Prize in 2029)

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Schrödinger: wave equation

Erwin Schrödinger

Oct 1925: Schrödinger read the Einstein paper on the latest developments in atomic physics that mentioned the de Broglie thesis

Presented a colloquium on the de Broglie ideas

Peter Debye's reaction: if matter is a wave, there must be a wave equation!
An equation must exist that describes the motion of a wave!

Erwin Schrödinger
1887 -1961

In a few months, Schrödinger derived the equation that is the foundation of quantum mechanics

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Schrödinger: wave equation

Requirements for the matter wave equation:

1. Consistent with de Broglie-Einstein postulates: $\lambda = \frac{h}{p} \qquad \nu = \frac{E}{h}$
2. Consistent with the energy equation: $E = \frac{mv^2}{2} + V(x,t) = \frac{p^2}{2m} + V(x,t) \qquad F = -\frac{\partial V}{\partial x}$
3. Must be linear in $\Psi(x,t)$, i.e. if $\Psi_1(x,t)$ and $\Psi_2(x,t)$ are solutions, then any linear combination must be a solution: $\Psi(x,t) = c_1\Psi_1(x,t) + c_2\Psi_2(x,t)$
Required to produce constructive/destructive interference

4. For a constant potential V, the solution must include a sinusoidal traveling wave of constant ν and λ
i.e., no external forces → constant momentum (wavelength), and energy (frequency)

For the logic behind the derivation of TDSE, see the YouTube Physics Explained lecture:
https://youtu.be/2WPA1L9uJqo

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Schrödinger: wave equation

The Schrödinger Equation (1926) (Nobel in 1933)

$$-\frac{\hbar^2}{2m}\frac{\partial^2\Psi(x,t)}{\partial x^2} + V(x,t)\Psi(x,t) = i\hbar\frac{\partial\Psi(x,t)}{\partial t}$$

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Wavefunction

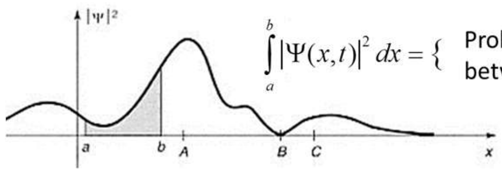
But what is the wavefunction $\Psi(x,t)$?

- It is a complex function, but measurement results are real numbers!

Max Born
1882-1970

Max Born, 1926 (Nobel 1954):
$|\Psi(x,y)|^2 dx$ represents the probability of finding particle between x and x+dx:

$$P(x,t)\,dx = |\Psi(x,t)|^2\,dx = \Psi^*(x,t)\,\Psi(x,t)\,dx$$



$$\int_a^b |\Psi(x,t)|^2\,dx = \left\{ \begin{array}{l} \text{Probability of finding the particle} \\ \text{between } a \text{ and } b, \text{ at time } t. \end{array} \right\}$$

Note: A particle with a definite energy has a definite frequency, so it is spread over infinite space
Note: If a particle has a specific location, it must be represented as a wave packet
→ You cannot predict with certainty the outcome of an experiment
→ Quantum mechanics offers statistical information about possible results

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Solving the Schrödinger Equation: time-independent V

$$i\hbar\frac{\partial\Psi}{\partial t} = -\frac{\hbar^2}{2m}\frac{\partial^2\Psi}{\partial x^2} + V(x)\Psi$$

Separate variables: $\Psi(x,t) = \psi(x)\varphi(t) \longrightarrow \begin{cases} \dfrac{\partial\Psi(x,t)}{\partial t} = \psi(x)\dfrac{d}{dt}\varphi(t) \\ \dfrac{\partial^2\Psi(x,t)}{\partial x^2} = \dfrac{d^2\psi(x)}{dx^2}\varphi(t) \end{cases}$

$$i\hbar\frac{1}{\varphi}\frac{d\varphi}{dt} = -\frac{\hbar^2}{2m}\frac{1}{\psi}\frac{d^2\psi}{dx^2} + V$$

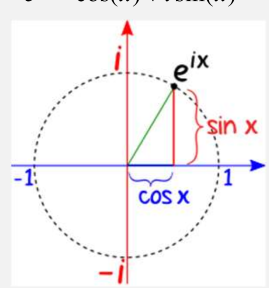
Function of t = Function of x for all x and t ⟶ Must be constant!

$$\begin{cases} i\hbar\dfrac{1}{\varphi}\dfrac{d\varphi}{dt} = E & \longrightarrow \boxed{\dfrac{d\varphi}{dt} = -\dfrac{iE}{\hbar}\varphi} \longrightarrow \boxed{\varphi(t) = e^{-iEt/\hbar}} \\ -\dfrac{\hbar^2}{2m}\dfrac{1}{\psi}\dfrac{d^2\psi}{dx^2} + V = E & \longrightarrow \boxed{-\dfrac{\hbar^2}{2m}\dfrac{d^2\psi}{dx^2} + V\psi = E\psi} \end{cases}$$

Time-independent Schrödinger equation

Euler's formula

$$e^{ix} = \cos(x) + i\sin(x)$$



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Solving the Schrödinger Equation: time-independent V

$$\boxed{-\frac{\hbar^2}{2m}\frac{d^2\psi}{dx^2}+V\psi=E\psi} \qquad \boxed{\varphi(t)=e^{-iEt/\hbar}}$$

Solutions: $\Psi(x,t)=\psi(x)\varphi(t)=\psi(x)e^{-iEt/\hbar}$

Note: $\int\Psi^*(x,t)\Psi(x,t)dx=\int\psi^*(x)e^{iEt/\hbar}\psi(x)e^{-iEt/\hbar}dx=\int\psi^*(x)\psi(x)dx$

Or: $\langle\Psi(x,t)|\Psi(x,t)\rangle\equiv\langle\psi(x)|\psi(x)\rangle$ The probability to find a particle at a certain location does not depend on time

We only need to solve for $\psi(x)$: $-\frac{\hbar^2}{2m}\frac{d^2\psi}{dx^2}+V\psi=E\psi$

We can have more than 1 solution: a set of $\psi_i(x)$ and respective E_i

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Solving the Schrödinger Equation: time-independent V

$$\Psi(x,t)=\psi(x)\varphi(t)=\psi(x)e^{-iEt/\hbar}$$

We only need to solve for $\psi(x)$: $-\frac{\hbar^2}{2m}\frac{d^2\psi}{dx^2}+V\psi=E\psi$

Can have more than 1 solution: a set of $\psi_i(x)$ and respective E_i

Notes:

- These are *stationary* solutions, the $|\Psi_i(x,t)|^2$ does not change in time
- These are states with *definite* energy
- ψ_i and E_i are *eigenfunctions* (or *eigenstates*) and *eigenvalues* of the time-independent Schrödinger equation
- These states are mutually orthogonal, i.e. $\int\psi_m(x)^*\psi_n(x)dx=\delta_{mn}$
- Any linear combination of these wavefunctions satisfies the Shrödinger equation:

$$\Psi(x,t)=\sum_i c_i\psi_i(x)e^{-iE_it/\hbar}$$

Expectation energy value $E=\sum_i|c_i|^2E_i$ But the energy of this state is not definite! The measurement will return one of the E_i values with the probability $|c_i|^2$

- It is a *complete* set: any solution to SE can be represented by such a linear combination

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Solving the Schrödinger Equation: free particle

Example: Free particle, V = 0

$$-\frac{\hbar^2}{2m}\frac{d^2\psi}{dx^2}=E\psi \qquad \text{Solution:} \quad \psi(x)=Ae^{ikx}+Be^{-ikx} \qquad E=\frac{\hbar^2k^2}{2m}$$

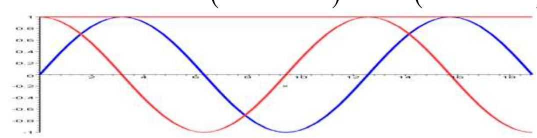
Combine with time-dependent part $\varphi(t)=e^{-iEt/\hbar}$

$$\Psi(x,t)=\underset{\rightarrow}{Ae^{i(kx-Et/\hbar)}}+\underset{\leftarrow}{Be^{-i(kx-Et/\hbar)}}$$

These are waves traveling in opposite directions, combine:

$$\Psi_k(x,t)=Ae^{i(kx-Et/\hbar)}$$

$$e^{i(kx-Et/\hbar)}=\cos(kx-Et/\hbar)+i\sin(kx-Et/\hbar)$$



$$k\equiv\pm\frac{\sqrt{2mE_k}}{\hbar},\ \text{with}\ \begin{cases}k>0\Rightarrow\ \text{traveling to the right}\\ k<0\Rightarrow\ \text{traveling to the left}\end{cases}$$

The energy is not quantized and can take any value, continuous spectrum

What is the amplitude A=?

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Particle as a wave packet

Any function can be represented as Fourier series integral

$$f(x)=\int_{-\infty}^{+\infty}g(k)e^{ikx}dk=\int_{-\infty}^{+\infty}\left[A(k)\cos(kx)+iB(k)\sin(kx)\right]dk$$

$$\Psi(x,0)=\frac{1}{\sqrt{2\pi}}\int_{-\infty}^{+\infty}\phi(k)e^{ikx}dk \qquad \Psi(x,t)=\frac{1}{\sqrt{2\pi}}\int_{-\infty}^{+\infty}\phi(k)e^{i\left(kx-\frac{\hbar k^2}{2m}t\right)}dk$$

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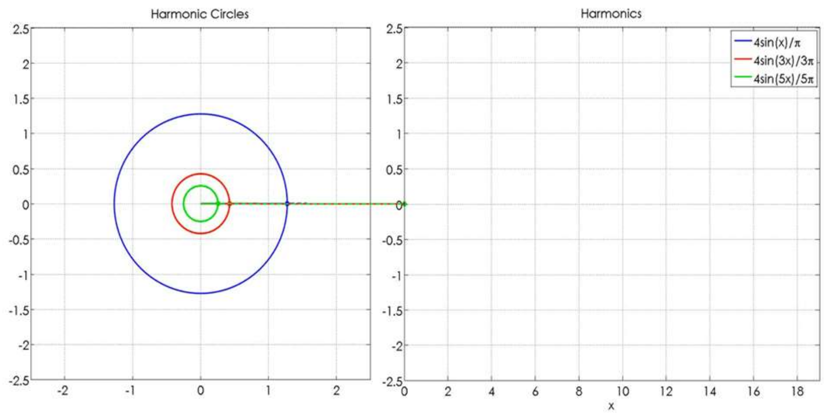
The Fourier series



Create your own: https://www.geogebra.org/m/aenhzjpz

Cool visualization: https://www.youtube.com/watch?v=r6sGWTCMz2k

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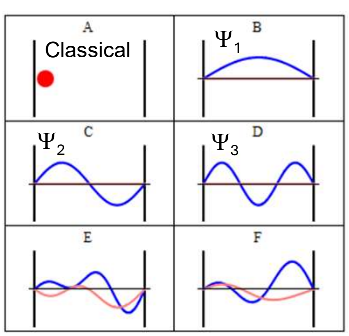
Particle in a box

Example: particle in a box $-\frac{\hbar^2}{2m}\frac{d^2\psi}{dx^2}+V\psi=E\psi$

Solutions: $\Psi_n(x,t)=\psi(x)\varphi(t)=\sqrt{\frac{2}{a}}\sin\left(\frac{n\pi}{a}x\right)e^{-iE_nt/\hbar}$ $\quad E_n=n^2\frac{\pi^2\hbar^2}{2ma^2}$

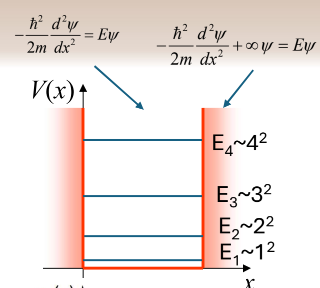
Standing waves! Discrete spectrum of energies (energy is quantized)

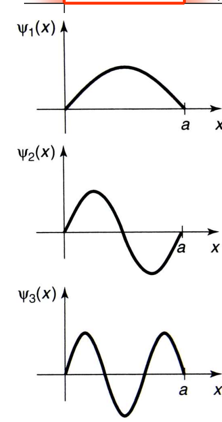


Re(Ψ)
Im(Ψ)

Stationary

Superposition of
Ψ_1, Ψ_2, Ψ_3 and Ψ_4,





http://en.wikipedia.org/wiki/Particle_in_a_box

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Particle in harmonic potential

$$-\frac{\hbar^2}{2m}\frac{d^2\psi}{dx^2}+\frac{1}{2}m\omega^2x^2\psi=E_n\psi$$

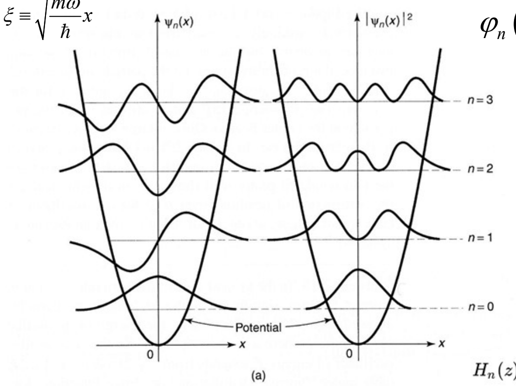
Note: any potential function V(x) near minimum can be approximated as V(x)~x^2 (the first =term in Taylor series, V(x)=$c_0+c_2x^2+c_3x^4+\ldots$)
Spring-mass: V = $kx^2/2=m\omega^2x^2$ → Hook's law → F = -dV/dx = -kx

Hermite polynomials

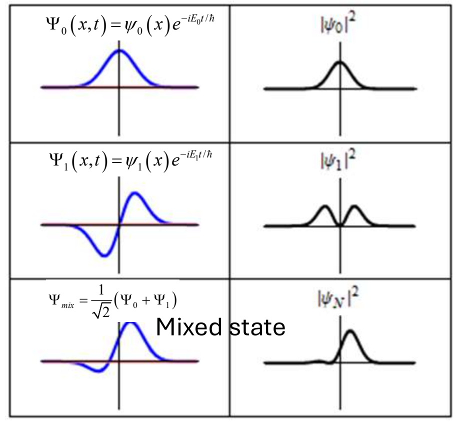
$$\psi_n(x)=\left(\frac{m\omega}{\pi h}\right)^{1/4}\frac{1}{\sqrt{2^n n!}}H_n(\xi)e^{-\xi^2/2}$$

$$\xi\equiv\sqrt{\frac{m\omega}{\hbar}}x$$

$$E_n=\hbar\omega(n+1/2)$$
$$n=0,1,2,\ldots$$
$$\varphi_n(t)=e^{-iE_nt/\hbar}$$



$$H_n(z)=(-1)^n e^{z^2}\frac{d^n}{dz^n}\left(e^{-z^2}\right)$$



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Particle wavepacket in a harmonic potential

Superposition:

$$\Psi(x,t)=\sum_{n=c-2whm}^{c+2fwhm}c_n\psi_n(x)e^{-iE_nt/\hbar}$$

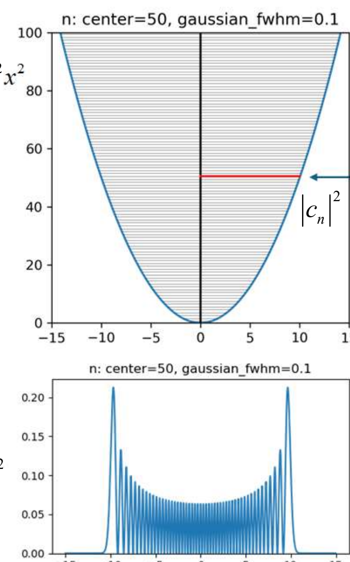
$$E_n=\hbar\omega(n+1/2)$$

Wavepacket center n_c = 50
Expansion coefficients follow a Gaussian distribution:

$$|c_n|^2=\frac{1}{\sigma\sqrt{2\pi}}\exp\left[-\frac{(n-n_c)^2}{2\sigma^2}\right]$$

$$fwhm=2\sqrt{2\ln 2}\sigma$$

$$V=\frac{1}{2}m\omega^2x^2$$



$$|\Psi(x,t)|^2$$

n=50, fwhm = 0
Only one level is populated
→stationary state

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Particle wavepacket in a harmonic potential

Superposition:

$$\Psi(x,t) = \sum_{n=c-2whm}^{c+2\,fwhm} c_n \psi_n(x) e^{-iE_n t/\hbar}$$

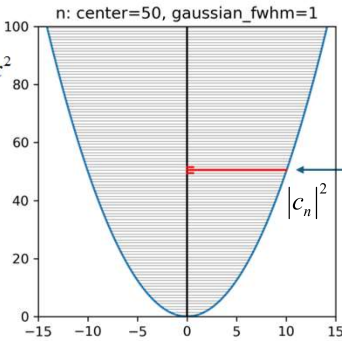
$$E_n = \hbar\omega(n+1/2)$$

Wavepacket center n_c = 50
Expansion coefficients follow a Gaussian distribution:

$$|c_n|^2 = \frac{1}{\sigma\sqrt{2\pi}} \exp\left[-\frac{(n-n_c)^2}{2\sigma^2}\right]$$

$$fwhm = 2\sqrt{2\ln 2}\,\sigma$$

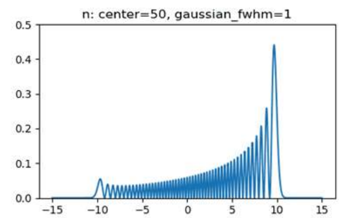
$$V = \frac{1}{2} m\omega^2 x^2$$



$|c_n|^2$

n=50, fwhm = 1
Two nearby states are slightly mixed in
Not a stationary state!

$$|\Psi(x,t)|^2$$



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Particle wavepacket in a harmonic potential

Superposition:

$$\Psi(x,t) = \sum_{n=c-2whm}^{c+2\,fwhm} c_n \psi_n(x) e^{-iE_n t/\hbar}$$

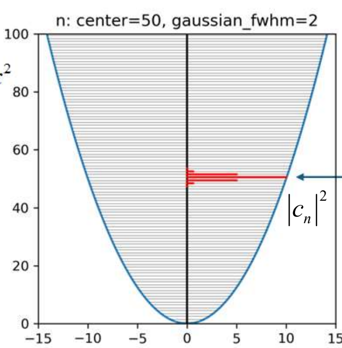
$$E_n = \hbar\omega(n+1/2)$$

Wavepacket center n_c = 50
Expansion coefficients follow a Gaussian distribution:

$$|c_n|^2 = \frac{1}{\sigma\sqrt{2\pi}} \exp\left[-\frac{(n-n_c)^2}{2\sigma^2}\right]$$

$$fwhm = 2\sqrt{2\ln 2}\,\sigma$$

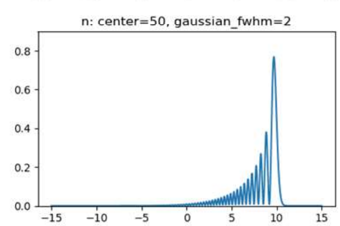
$$V = \frac{1}{2} m\omega^2 x^2$$



$|c_n|^2$

n=50, fwhm = 2
A few nearby states are mixed in
Not a stationary state!

$$|\Psi(x,t)|^2$$



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Particle wavepacket in a harmonic potential

Superposition:

$$\Psi(x,t) = \sum_{n=c-2whm}^{c+2fwhm} c_n \psi_n(x) e^{-iE_n t/\hbar}$$

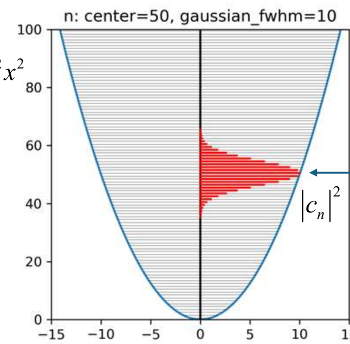
$$E_n = \hbar\omega(n + 1/2)$$

Wavepacket center $n_c = 50$
Expansion coefficients follow a Gaussian distribution:

$$|c_n|^2 = \frac{1}{\sigma\sqrt{2\pi}} \exp\left[-\frac{(n-n_c)^2}{2\sigma^2}\right]$$

$$fwhm = 2\sqrt{2\ln 2}\,\sigma$$

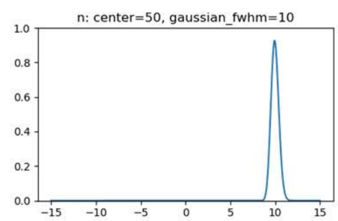
$$V = \frac{1}{2} m\omega^2 x^2$$



n=50, fwhm = 10

Several nearby states are mixed in
Not a stationary state!

$$|\Psi(x,t)|^2$$



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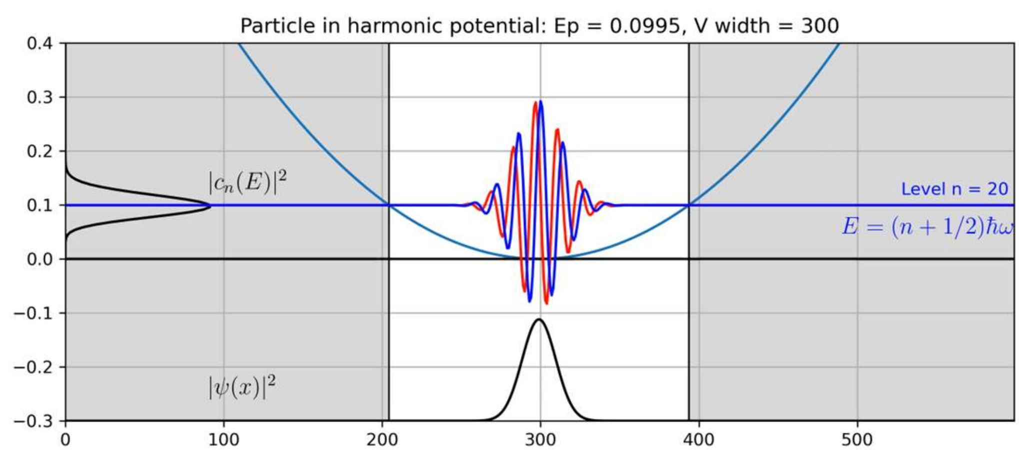
Particle in harmonic potential



Simulation by S. Savikhin

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Particle wavepacket in a harmonic potential

```python
import matplotlib.pyplot as plt
import numpy
import math
#Choose simple units
m=1.
w=1
hbar=1.
#Discretized space, x-axis
dx = 0.005
x_lim = 15
x = numpy.arange(-x_lim,x_lim,dx)
#wavepacket center and width
center = 50
width = 20
```

```python
# service functions
def hermite(x, n):
    xi = numpy.sqrt(m*w/hbar)*x
    herm_coeffs = numpy.zeros(n+1)
    herm_coeffs[n] = 1
    return numpy.polynomial.hermite.hermval(xi, herm_coeffs)
def stationary_state(x,n):
    xi = numpy.sqrt(m*w/hbar)*x
    prefactor = 1./math.sqrt(2.**n * math.factorial(n)) * (m*w/(numpy.pi*hbar))**(0.25)
    psi = prefactor * numpy.exp(- xi**2 / 2) * hermite(x,n)
    return psi
def gauss(pos,center,width):
    sigma = width/2.35492
    return 1./sigma/math.sqrt(2*math.pi)*math.exp(-(pos-center)**2/sigma**2/2)
```

```python
# make superposition
i=1 # output figure file name counter
for t in numpy.arange(0,2*math.pi,math.pi/15):
    wf_re = x*0
    wf_im = x*0
    for n in range(center-width*2,center+width*2):
        phase = (n+0.5)*t  # beats between n and n+1 define the period, i.e. it is omega
        wf_re = wf_re + math.cos(phase)*stationary_state(x,n)*math.sqrt(gauss(n,center,width))
        wf_im = wf_im + math.sin(phase)*stationary_state(x,n)*math.sqrt(gauss(n,center,width))
    plt.figure(figsize=(5, 3))
    plt.ylim(0,1)
    plt.plot(x,wf_re**2 + wf_im**2) #,label=f'center={center}, fwhm={width}')
    plt.title(f'n: center={center}, gaussian_fwhm={width}')
    plt.savefig(f'fig-c{center:002}-w{width:002}-{i:003}', dpi = 100, facecolor = 'white')
    i = i + 1
```

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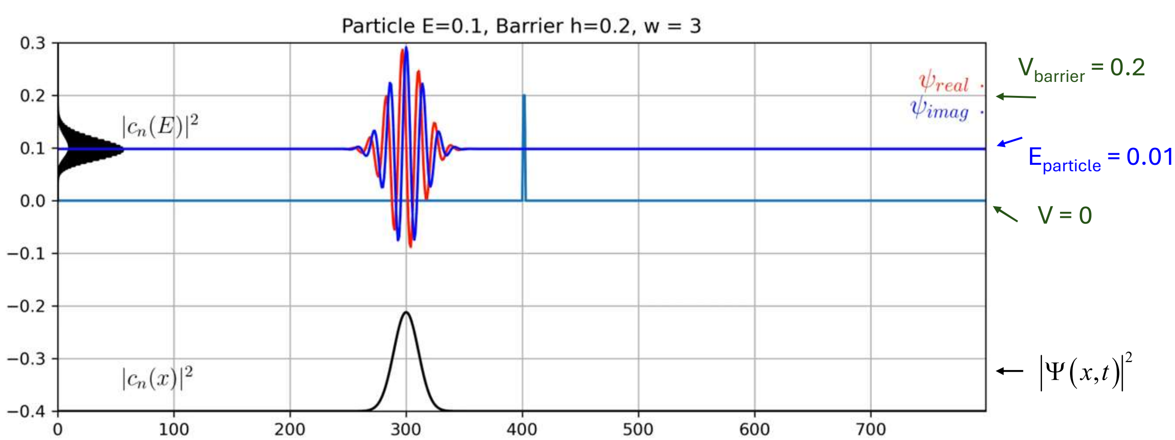
Quantum mechanical tunneling



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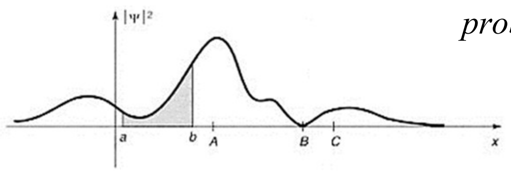
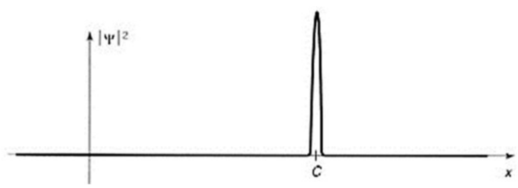
Measurement of a position



$$probability[a...b]=\int_a^b \varphi^*(x)\varphi(x)dx$$

What will be the result of the position measurement?

→ You will get only 1 number, uncertainty in the measurement



Suppose you got *c*, and then take another measurement on the same particle. What will you get?

You will get c again!
→ The wavefunction will collapse into a definite position state!

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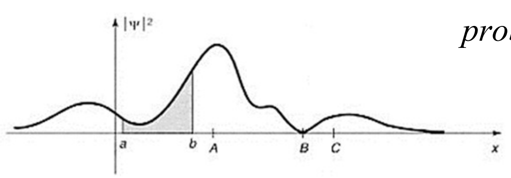
Measurement and expectation values



$$probability[a...b]=\int_a^b \varphi^*(x)\varphi(x)dx$$

What will be the *average* of many position measurements on a particle with such a wavefunction?

$$\langle x\rangle=\int_{-\infty}^{\infty} x\cdot P(x)dx=\int_{-\infty}^{\infty} x\varphi^*(x)\varphi(x)dx=\int_{-\infty}^{\infty}\varphi^*(x)x\varphi(x)dx$$

Expectation value: $\langle x\rangle=\int_{-\infty}^{\infty}\varphi^*(x)x\varphi(x)dx\equiv\langle\varphi|x|\varphi\rangle$ ← Dirac's **Bra-ket** notation

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Operators, measurements, and expectation values

Physical quantities are represented by operators:

Expectation value: $\langle A \rangle = \left\langle \Psi \middle| \hat{A} \middle| \Psi \right\rangle \equiv \int \Psi^* \hat{A} \Psi dr$

Examples:

Operators:

$\langle x \rangle = \int \Psi^* \cdot (x) \cdot \Psi dx$ $\quad (x)$ - position

$\langle p \rangle = \int \Psi^* \cdot \left(\frac{\hbar}{i} \frac{\partial}{\partial x} \right) \cdot \Psi dx$ $\quad \hat{p}_x = \left(\frac{\hbar}{i} \frac{\partial}{\partial x} \right)$ - momentum

$\langle E \rangle = \int \Psi^* \hat{H} \Psi dx$ $\quad \hat{H} = -\frac{\hbar^2}{2m} \frac{\partial^2}{\partial x^2} + V$ energy

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Operators, eigenfunctions, and eigenvalues

$$\langle A \rangle = \langle \varphi | \hat{A} | \varphi \rangle$$
$$\hat{X} = x$$
$$\hat{H} = -\frac{\hbar^2}{2m} \frac{d^2}{dx^2} + V$$
$$\hat{p} = -i\hbar \frac{d}{dx}$$

For any operator, we can find a set of wavefunctions that will produce a definite result in a respective measurement.
Solve the equation: $\hat{A}\varphi = A\varphi$

The solutions are called *eigenfunctions* φ_i and *eigenvalues* A_i of operator $\hat{A}$

Note: operator A acting on the eigenfunction just multiplies it by some constant (eigenvalue)

These solutions are complete, orthogonal and normalized *(orthonormalized):*

$$\langle \varphi_i | \varphi_j \rangle = \delta_{ij}$$

Any other possible wavefunction can be represented as a linear combination of these solutions, or represented in *the basis of eigenfunctions of operator* $\hat{A}$

$$\Psi = \sum_i c_i \varphi_i$$

The same wavefunction can be represented as a superposition of eigenfunctions of a different operator (i.e., using a different set of eigenfunctions), though the expansion coefficients c_i (*eigenvector*) will be different.

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Vector representation of a wavefunction

Time-Independent Schrödinger Equation (TISE) $-\frac{\hbar^2}{2m}\frac{d^2\psi}{dx^2} + V\psi = E\psi$ Or: $\hat{H}\psi_i = E\psi_i$

$\hat{H} = -\frac{\hbar^2}{2m}\frac{d^2}{dx^2} + V$ Energy operator (Hamiltonian)

Then any other possible wavefunction can be represented as : $\Psi(x,t) = \sum_i c_i \psi_i(x) e^{-iE_i t/\hbar}$

The expansion coefficients c_i uniquely define this wavefunction *in the basis of eigenfunctions of H*.

Can represent the wavefunction as a vector in the basis of this operator $\Psi = \begin{pmatrix} c_1 \\ c_2 \\ c_3 \\ \dots \end{pmatrix}$

Normalization: $\sum |c_i|^2 = 1$

$\psi_i = \begin{pmatrix} 0 \\ \dots \\ c_i = 1 \\ \dots \end{pmatrix}$

Note: the vector components will depend on the choice of the *basis,* i.e., in bases of eigenfunctions of different operators, the coefficients will be different, but they will represent *the same wavefunction!*

This is similar to expressing a geometrical vector in different systems of coordinates.
You can think of the coefficients c_i as of the projections of the wavefunction Ψ to eigenfunctions ψ_i

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Matrix representation of operators

Let (ψ_i, E_i) be the solutions of: $\hat{H}\psi_i = E_i\psi_i$

$\Psi = \begin{pmatrix} c_1 \\ c_2 \\ c_3 \\ \dots \end{pmatrix}$

The Hamiltonian in the basis of its own eigenfunctions can be represented as a diagonal matrix with respective eigenvalues on its diagonal:

$$\mathbf{H} = \begin{pmatrix} E_1 & 0 & 0 & \dots \\ 0 & E_2 & 0 & \dots \\ 0 & 0 & E_3 & \dots \\ \dots & \dots & \dots & \dots \end{pmatrix}$$

The expectation value for energy then can be found in the matrix form as:

$$\langle H \rangle = \langle \psi | \hat{H} | \psi \rangle = \boldsymbol{\psi}^{\mathrm{T}}\mathbf{H}\boldsymbol{\psi} = \begin{pmatrix} c_1^* & c_2^* & c_3^* & \dots \end{pmatrix} \begin{pmatrix} E_1 & 0 & 0 & \dots \\ 0 & E_2 & 0 & \dots \\ 0 & 0 & E_3 & \dots \\ \dots & \dots & \dots & \dots \end{pmatrix} \begin{pmatrix} c_1 \\ c_2 \\ c_3 \\ \dots \end{pmatrix} = \sum_i |c_i|^2 E_i$$

Note: in the basis of other operator eigenfunctions, the H-matrix will look different.
Note: as long as the wavefunction vector and operator matrix are expressed in the same basis, the $\langle A \rangle = \boldsymbol{\psi}^{\mathrm{T}}\mathbf{A}\boldsymbol{\psi}$

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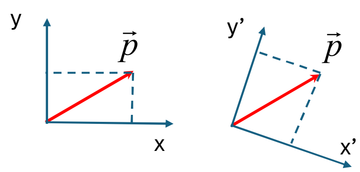
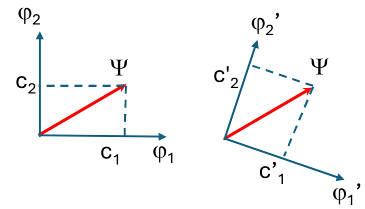
Vector: projections to basis functions

The wavefunction vector components and operator matrix components depend on the choice of the basis*!*
Iin bases of eigenfunctions of different operators, the coefficients will be different, but they will represent *the same wavefunction!*

This is similar to expressing a geometrical vector in different systems of coordinates.
You can think of the coefficients c_i as the projections of the wavefunction Ψ to eigenfunctions ψ_i





$$\vec{p} = x\hat{i} + y\hat{j} = x'\hat{i}' + y'\hat{j}'$$

Projections:

$$\vec{p}\cdot\hat{i} = x\hat{i}\cdot\hat{i} + y\hat{j}\cdot\hat{i} = x$$

$$\vec{p}\cdot\hat{j} = y$$

$$\vec{p}\cdot\hat{i}' = x'$$

$$\vec{p}\cdot\hat{j}' = y'$$

$$\Psi = c_1\varphi_1 + c_2\varphi_2 = c'_1\varphi'_1 + c'_2\varphi'_2$$

$$\langle\Psi|\varphi_1\rangle = \langle c_1\varphi_1 + c_2\varphi_2|\varphi_1\rangle = c_1\langle\varphi_1|\varphi_1\rangle + c_2\langle\varphi_2|\varphi_1\rangle = c_1$$

$$\langle\Psi|\varphi_2\rangle = \langle c_1\varphi_1 + c_2\varphi_2|\varphi_2\rangle =_1 \langle\varphi_1|\varphi_2\rangle + c_2\langle\varphi_2|\varphi_2\rangle = c_2$$

What is the length of this vector?

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